Pneumorrhachis: A potential and under-recognised pitfall for thoracic epidural analgesia in blunt chest trauma

Dear Editor,

Pneumorrhachis, the presence of air within the spinal canal, is increasingly recognised as part of the spectrum of barotrauma associated with blunt thoracic injury. Pneumorrhachis is an uncommon finding, reported in approximately 5.8% of patients with pneumomediastinum on computed tomography (CT) imaging.[1] Although epidural pneumorrhachis is usually considered a benign and self-limiting radiological finding, its presence in the intradural space may still have important implications for neuraxial procedures.[2] The underlying mechanism is attributed to a rise in intrathoracic pressure following blunt chest trauma, leading to alveolar rupture and air dissection along the perivascular interstitium (Macklin effect). The escaped air tracks along fascial planes into the mediastinum and may further extend into the spinal canal through the intervertebral neural foramina, facilitated by the absence of firm anatomical barriers.[3] While pneumorrhachis is a recognised radiological entity, its implications for neuraxial decision-making in trauma patients remain under-reported. We highlight the relevance of pneumorrhachis as a potential decision-making challenge when planning thoracic epidural analgesia (TEA) in patients with bilateral multiple rib fractures.

A 45-year-old male presented following blunt thoracic trauma with respiratory distress. On presentation, the patient had tachypnoea with an increased respiratory rate of 30 to 35 per minute, with an oxygen saturation of 94% on non-rebreathing mask with a heart rate of 115 to 120 per minute and a blood pressure of 122/52 mmHg. Contrast-enhanced CT of the chest and abdomen revealed extensive subcutaneous emphysema extending from the neck to the inguinal region, pneumomediastinum, bilateral hydro-pneumothorax requiring thoracostomy, and pneumorrhachis involving the dorsal epidural space (C3–T1) without evidence of tracheobronchial, oesophageal, or spinal column injury. Additionally, there were multiple rib fractures involving the left 2nd–7th ribs and right 6th–8th ribs, along with a fracture of the manubrium. The patient remained neurologically intact [Figure 1a, b] and reported severe pain (numerical rating score 8/10) with splinting and reduced inspiratory effort. He subsequently developed worsening respiratory distress with increasing respiratory rate and declining oxygenation, necessitating endotracheal intubation and invasive mechanical ventilation.

TEA is often favoured in patients with multiple rib fractures and flail chest for optimal analgesia and improvement in pulmonary mechanics.[4] However, the presence of pneumorrhachis introduces important technical, neurological, and medicolegal considerations that may influence decision-making for TEA [Table 1]. Meticulous documentation of baseline neurological status is essential prior to any neuraxial intervention. In addition, the presence of pneumomediastinum represents a relative concern when considering neuraxial techniques, particularly in the context of associated pneumorrhachis.

A cautious and individualised approach is warranted in this context. Temporarily deferring TEA may be

considered on a case-by-case basis as pneumorrhachis is often self-limiting with spontaneous resorption of intraspinal air; however, the optimal timing of neuraxial intervention remains uncertain.[2] If epidural analgesia is undertaken, use of loss of resistance to saline, minimisation of multiple attempts, and incremental dosing may be considered to enhance procedural safety. However, the presence of intraspinal air may alter epidural pressure dynamics, potentially affecting the reliability of the loss of resistance technique and leading to false identification of the epidural space. Alternatively, fascial plane techniques such as erector spinae plane block, paravertebral blocks, intercostal nerve blocks, serratus anterior plane block, and pectoral nerve I/II blocks may provide effective analgesia while avoiding neuraxial instrumentation. Vigilance for local anaesthetic systemic toxicity remains essential when managing extensive bilateral injuries requiring higher volumes. A multimodal analgesic strategy incorporating intravenous opioids, non-steroidal anti-inflammatory drugs, and adjuncts such as ketamine can provide effective pain control while preserving neurological assessment and avoiding neuraxial risks.

We opted for systemic multimodal analgesia in our patient, prioritising procedural safety and preservation of neurological assessment. This included intravenous fentanyl, paracetamol, and ketorolac, with adjunctive use of titrated low-dose ketamine as required. This approach provided effective analgesia while allowing ongoing neurological monitoring. The patient demonstrated gradual improvement in respiratory status with supportive care.

While pneumorrhachis is often considered a radiological curiosity, its presence may influence critical analgesic decisions in trauma care. Recognising pneumorrhachis as a potential pitfall for neuraxial techniques may help anaesthesiologists avoid technical failure and diagnostic uncertainty in an already complex clinical setting.

Declaration of Patient consent

The authors certify that they have obtained all appropriate patient consent forms. In the form, the patient consented to the images and other clinical information to be reported in the journal. The patient understands that his name and initials will not be published and due efforts will be made to conceal his identity, but anonymity cannot be guaranteed.

Disclosure of use of artificial intelligence (AI)-assistive or generative tools

Artificial intelligence (AI) tools have not been utilised in the manuscript, except that software has been used for grammar corrections and references.

Author contributions

RN: Conceptualisation, Writing – original draft; AG: Literature review, Conceptualisation, Writing – original draft; SA: Supervision, Critical revision of manuscript; KJ: Supervision, Writing – review and editing, Critical revision of manuscript.

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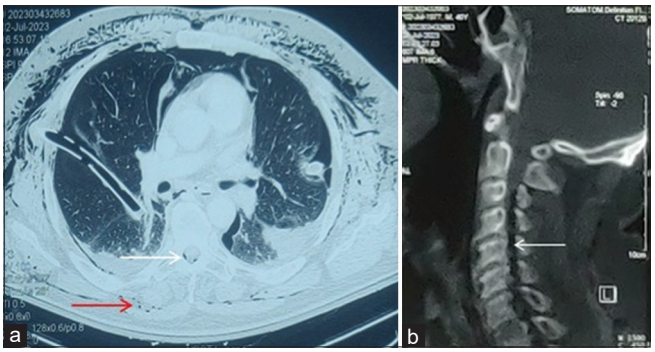


Figure 1: Computed tomography images demonstrating pneumorrhachis. (a) Axial CT chest image showing air within the spinal canal (white arrow) with air dissecting along the fascial planes (red arrow). (b) Sagittal CT of the cervical spine demonstrating air in the dorsal spinal canal (white arrow) extending from C3 to T1. CT: Computed tomography

Table 1: Challenges of thoracic epidural analgesia in the presence of pneumorrhachis

Domain	Challenge	Clinical consequence
Technical	False loss of resistance due to epidural air[5]	Misidentification of epidural space and catheter malposition[5]
Drug spread	Air within the epidural space acting as a physical barrier[5]	Patchy or inadequate analgesia, failure to meet high analgesic demands[5]
Neurological	Increased intraspinal pressure or progression of pneumorrhachis[6]	Potential spinal cord or nerve root compression (theoretical)[6]
Diagnostic	New neurological deficits following epidural placement[6]	Difficulty distinguishing pneumorrhachis progression from procedural complications and progression of pneumomediastinum following epidural placement with important medicolegal implications[6]

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